

learning-surface

Contract — The learning surface

Routes

`/api/areas/{id}/lessons · /lessons/{lesson} · /lessons/{lesson}/state
· /materials · /assessment · /assessment/submit`

All six require the opaque X-Session header. One channel only — a sibling module accepts a session three ways, and three places to supply it are three places for them to disagree.

The client must send it. It did not, and every area page rendered an error card: 400 without, 200 with. The header builder is **one exported function** consumed by every client.

Lessons carry content

Every served lesson of a published area carries a non-empty **body** — declared **upstream in the kit**, kept distinct from summary. Collapsing them was the semantic lie that caused 339 bodies to be discarded on every build.

Measured: 318 of 318 served, median 328 words, longest 4,563.
Twelve lessons under 100 words are **short, not padded**.

The assessment gate

While locked: **zero** prompts, options and answers, with the remaining requirement stated and `pass_percent` shown **before** the attempt.

Enforced **by construction**: the taking projection is a map with **no answer field at all**, so no future upstream field can leak by defaulting to present.

And the lock is not bypassable one route over. The practice deck and the graded bank are the same authored rows — 224 of 224 match by prompt, 0 of 224 by id — so gating practice would delete a feature the contract intends. Graded rows are **withheld from the open route, and the withholding says so.**

Fail-closed: nil, unreadable **or stale** catalog withholds every key. A graded prompt the snapshot never saw would look “graded nowhere” to a literal index and be published.

Absence must be true

An area with no assessment states **only what the server established**. It may not claim “there is genuinely none” unless that has been determined — the snapshot may only license what it holds.

Measured: 27 of 39 areas serve a test, 205 questions; 12 carry none and each says so with its evidence count.

Submission validation

responses is required and must be an array. A misspelled top-level key is refused **400 naming the offending key**, before the store is touched — it previously returned 200 and permanently recorded a 0% attempt. **responses: [] still grades**: a blank paper is a real submission.

Asserted by

verify-lesson-content.sh (two arms: in-process **and live**) · prover 9 mutations · verify-bank-reachability.sh + prover 9 · verify-answer-key-disclosure.sh + prover **16** · verify-submission-validation.sh + prover **11**. Each carries a vacuity refusal exiting 2 and **overreach mutations** — a practice-only question stripped of its answer, a valid submission refused — which a gate written only against the leak would have certified as correct.